

Data-Driven Cutting Torque Modelling of Pedicle Screw Channel Creation for Haptic Surgical Simulator

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INTRODUCTION

Pedicle screws are commonly used in spinal fusion surgery for their high-strength fixation and improved correction performance. However, it has a high risk of complications (i.e., 7-30%), resulting in neurological impairments and organ breaches [1], [2]. Thus, extensive training is needed to minimize risks. A haptic simulator can offer a more flexible, lower risk, and highly repeatable training medium rather than conventional training methods (e.g. cadavers). A typical bone drilling simulator consists of a motorized axis that simulates the reaction torque that must be modeled as it is generated during bone drilling.

Previous work on modeling the cutting torque includes the implementation of a physics-based drilling model [3]. However, this model is based on a high-speed twist drill, which is rarely used by surgeons due to the complex geometry of the pedicle bone region. Another approach models a manual surgical probe using polynomial regression derived from constant thrust force experiments [4]. However, this approach introduces diverse drilling models based on tool geometry zones, which limits the generalization between operating conditions and material properties, posing a challenge for real-time deployment. In this work, We propose a data-driven cutting-torque model for pedicle channel creation that avoids geometry zones, creating a general model across varying operating parameters with low-RPM manual hand-drilling conditions.

MATERIALS AND METHODS

The experimental apparatus is constructed as shown in Fig. 1a, with a CNC machine (Doosan Mynx 6500, DN Solutions) integrated with multi-modal sensors (3,5,6 in Fig. 1a). In this drilling experiment, a standard surgical probe (Oury Medical D11438M, Oury Medical) was used with a unique geometry detailed in Fig. 1b, which

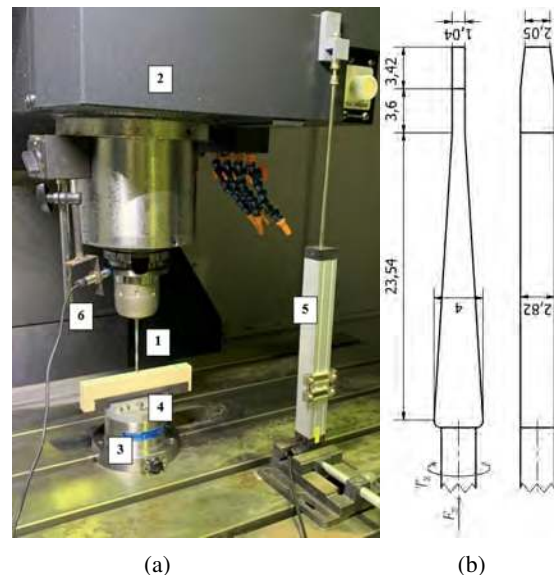


Fig. 1 (a) Overview of experimental apparatus; 1) Surgical probe, 2) CNC machine, 3) Force and Torque sensor, 4) Fixture, 5) Displacement sensor, 6) Proximity sensor. (b) Geometry of the Prism-shaped tip surgical probe

is identical to clinical practice. Polyurethane foam (Block Solid Foam, SAWBONES) with standardized densities of 10, 15, 20 and 25 PCF (pounds per cubic foot) was used to simulate low to medium-high cancellous bone density [5]. During the drilling process, the axial force (F_z) and the cutting torque (T_z) were measured with a force & torque sensor (Kistler 9272, Kistler) mounted below the specimen fixture with a sampling rate of 50 Hz.

The drilling parameters were selected to match the motion of the surgeon's hand during manual drilling, with low-RPM spindle speeds (10, 16, 25, 40, 63, 100 RPM) and feed rates (10, 15, 20 mm/min). A 31 mm drill bit insertion was conducted with a total of 198

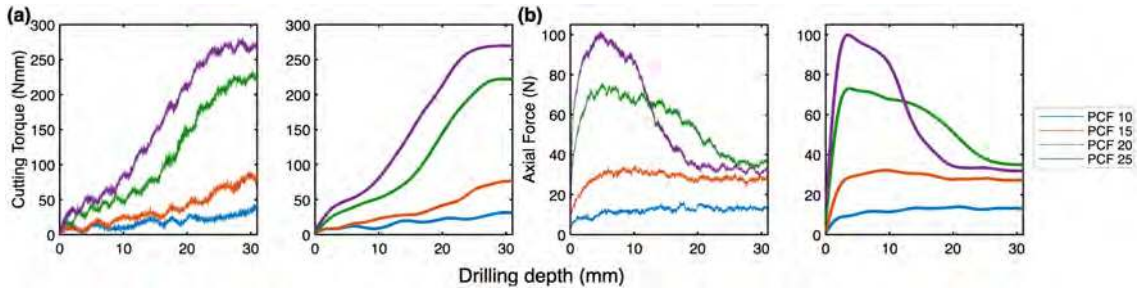


Fig. 2 (a) Cutting torque as a function of drilling depth showing unfiltered signals (left) and filtered signals (right). (b) Axial force as a function of drilling depth showing unfiltered signals (left) and filtered signals (right). While the unfiltered signals contain the full drilling dynamics, the filtered signals preserve the dominant features for modeling.

TABLE I Correlation of Input Parameters with Torque

Parameter	Density	Spindle Speed	Depth	Feed Rate	Thrust Force
Correlation	0.57	-0.28	0.51	0.08	0.76

experiments (for each parameter, 3 trials for 10, 15 and 20 PCF; and 2 trials for 25 PCF). A fourth-order Butterworth low-pass filter is used to filter the data with a cutoff frequency of 10 Hz, as the dominant dynamics lie below this frequency due to the low-RPM spindle speed (i.e. 100 RPM $\approx$ 1.67 Hz). Three machine learning (ML) models were evaluated for their ability to capture high-order nonlinear relationships inherent in the drilling process, which are: a three-layer Multilayer Perceptron Neural Network (MLP NN) with 256, 128 and 64 nodes in each layer, a 7th order polynomial (Poly 7 Deg) model, and a K-Nearest Neighbor (KNN) model with a K value of 50. The data set was divided into 60% training, 20% validation, and 20% testing sets. To investigate the effect of each parameter on the cutting torque, a correlation matrix was used. The axial force, density, depth, spindle speed, and feed rate were used as inputs, and the cutting torque as output.

RESULTS

The experimental data show that material density is the dominant factor that influences the cutting torque. As shown in Fig. 2, a rapid increase in thrust force was observed during the initial penetration phase, which then stabilizes or decreases slightly in denser materials as the drill bit becomes fully engaged. The correlation matrix in Table I shows that torque is positively correlated with material density and feed rate, while negatively correlated with spindle speed. Table II summarizes the performance of ML models, where the Multilayer Perceptron (MLP) achieved the highest accuracy, successfully identifying the system dynamics, and can replicate the conditions of manual free-hand drilling.

DISCUSSION

The results of this study demonstrate how drilling parameters influence cutting torque and capture relationships using a data-driven model. Significant

TABLE II Comparison of ML Model Results

Model	Validation RMSE (Nmm)	Test RMSE (Nmm)	Validation R^2	Test R^2
MLP NN	7.846	6.339	0.999	0.998
Poly 7 Deg	7.397	7.946	0.999	0.997
KNN	5.784	8.783	0.999	0.996

variations in force and torque with material density suggest the potential to provide signals that may differentiate bone types by density when drilling. However, the current study only explored material densities equivalent to cancellous bones, overlooking additional dynamics associated with drilling across the cancellous-cortical transition and cortical bones. In addition, the experimental setup only partially replicates real surgical conditions, and further investigation is required to establish broader clinical relevance.

The cutting torque is influenced by the prism-shaped geometry of the surgical probe, resulting in an extended torque growth region in depths greater than motorized drill bits, which reach steady-state torque at shallower depths. Chip accumulation is observed within the drilled hole, as the surgical probe also lacked a flute for chip removal. This finding can introduce additional nonlinear effects in the drilling process that have not been investigated in this study. Future works will investigate drilling in cortical bone and across cancellous-cortical transition, capturing clinically relevant drilling dynamics. In addition, the effects on how the accumulation of chips affects the cutting torque and thrust force will be studied, as well as implementing harmonic drilling motions to replicate identical surgical drilling conditions.

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